RED OASIS | FIELD RESEARCH BRIEF

Sharifa • Oman pilot site, date-palm baseline, Mars greenhouse reference, and field-validation plan

PILOT SITE	Ad Dariz agricultural area, Wilayat Ibri, Al Dhahirah, Oman; Falaj al-Dariz paper is the historical regional reference.	**COORDINATES**	23.308639 N, 56.613083 E (team roadmap point; exact orchard unconfirmed)
CROP	Date palm (*Phoenix dactylifera*), dominant in the historical Dariz study.	**AREA**	Historic Falaj al-Dariz cropped area: 255.44 ha (MRMEWR 2001 GPS/GIS survey, as reported by Al-Ghafri). This is falaj-wide, not the current mapped grove boundary.
SATELLITE	Esri image dated 2024-05-08 shows cultivated plots around the point; marker is near a track/parcel edge. See separate marked image.	**LOCATION NOTE**	The 2007 map places al-Dariz near 23°19′N, 56°36′E; 56°47′E is al-Nujaid. Team point is about 1.45 km SW of the mapped village node; exact parcel remains unconfirmed.

1 | Earth crop and water baseline

Al-Ghafri's Falaj al-Dariz case study used observations from 2002–2003. Reported annual date-palm crop water demand (ETc) was **3,620 mm**; field supply including effective rainfall was **3,235 mm**. The demand/supply ratio was **1.12**: demand exceeded supply by 385 mm (about 10.6% of demand). Converted values are approximately **36,200 m3/ha/year** demanded versus **32,350 m3/ha/year** supplied. These are historical site-study values, not current measurements or a universal date-palm prescription. [1]

Fruit development follows flowering/pollination and five stages: **Hababouk** → **Kimri** → **Khalal** → **Rutab** → **Tamar**. Interpret irrigation alongside season, cultivar, palm age, soil, and irrigation method; confirm the current schedule with a grower before using it as an operational recommendation. [2]

2 | Mars mode: NASA evidence and prototype settings

NASA's MarsOASIS is a greenhouse concept for growing Outredgeous lettuce, with autonomous monitoring and a closed water-recycling system; it is not evidence that date palms have been grown on Mars. [3] NASA greenhouse research gives broad plant-growth bounds of about **10–35 C** and **40–85% relative humidity**. [4] For a reproducible demo profile, use a NASA ISS lettuce reference: **23.25 C, 16 h light / 8 h dark, 200 micromol m^-2 s^-1**. These are lettuce chamber settings, not validated Mars or date-palm setpoints. [5] Set water recovery ON; test external make-up water at **50%, 75%, and 100%** of the Earth baseline as team stress-test assumptions, not NASA-published Mars values.

3 | Published Oman comparison data

A 2015–2016 Oman field trial at Al Kamil and Al Wafi (published 2022) tested Khalas palms. Water use was **9,966.154 m3/ha/year** with bubbler irrigation vs **6,420.600 m3/ha/year** with subsurface drip at 60% ETc (35.6% less); yields were **79.0 vs 78.3 kg/palm** (no significant difference reported). At 40% and 20% ETc, yields fell to 69.8 and 68.8 kg/palm. These are comparator results from South Al Sharqiyah, not Dariz measurements. [6]

4 | Climate context and remaining verification

NASA POWER's 25-year (2001–2025) climatology at the roadmap point gives mean 2 m temperature 27.75 C/year (17.58 C in January; 35.76 C in July) and mean all-sky solar radiation 22.03 MJ/m2/day. The official Ibri weather station recorded 176.4 mm total rainfall in 2025; its monthly mean temperature ranged from 19.3 C (January) to 38.4 C (July). These are gridded/station context, not measurements at the grove. Exact orchard polygon/area, local irrigation practice, and thresholds still need local confirmation. A SMAP soil-moisture pixel is far larger than the orchard. [7–10]

REFERENCES

[1] Al-Ghafri, A. (2006). "Aflaj's Irrigation Water Demand/Supply Ratio: Two Case Studies." Journal of Agricultural and Marine Sciences, 11(1), 85–92.

[2] FAO. Date Palm: Botanical and Systematic Description — fruit growth stages. [3] NASA NTRS. Mars OASIS System Reference Manual.

[4] Wheeler (2000), NASA NTRS. Can CO2 be Used as a Pressurizing Gas for Mars Greenhouses?. [5] NASA OSDR, OSD-267, Veggie lettuce protocol.

[6] Dhehibi et al. (2022). Economic and environmental evaluation of different irrigation systems for date palm production in the GCC: Oman and Saudi Arabia. [7] NSIDC, SMAP L3 soil-moisture product.

[8] Al-Ghafri (2007), map locates al-Dariz at 23°19′N, 56°36′E and al-Nujaid at 23°28′N, 56°47′E; area from MRMEWR 2001 GPS/GIS survey. Conference paper. [9] NASA POWER, 25-year climatology API (Jan 2001–Dec 2025), T2M and all-sky solar radiation at the team point. [10] Oman NCSI, Ibri station rainfall and mean temperature, 2025. [11] Esri World Imagery, point-specific Vantor/WorldView-3 image dated 2024-05-08.